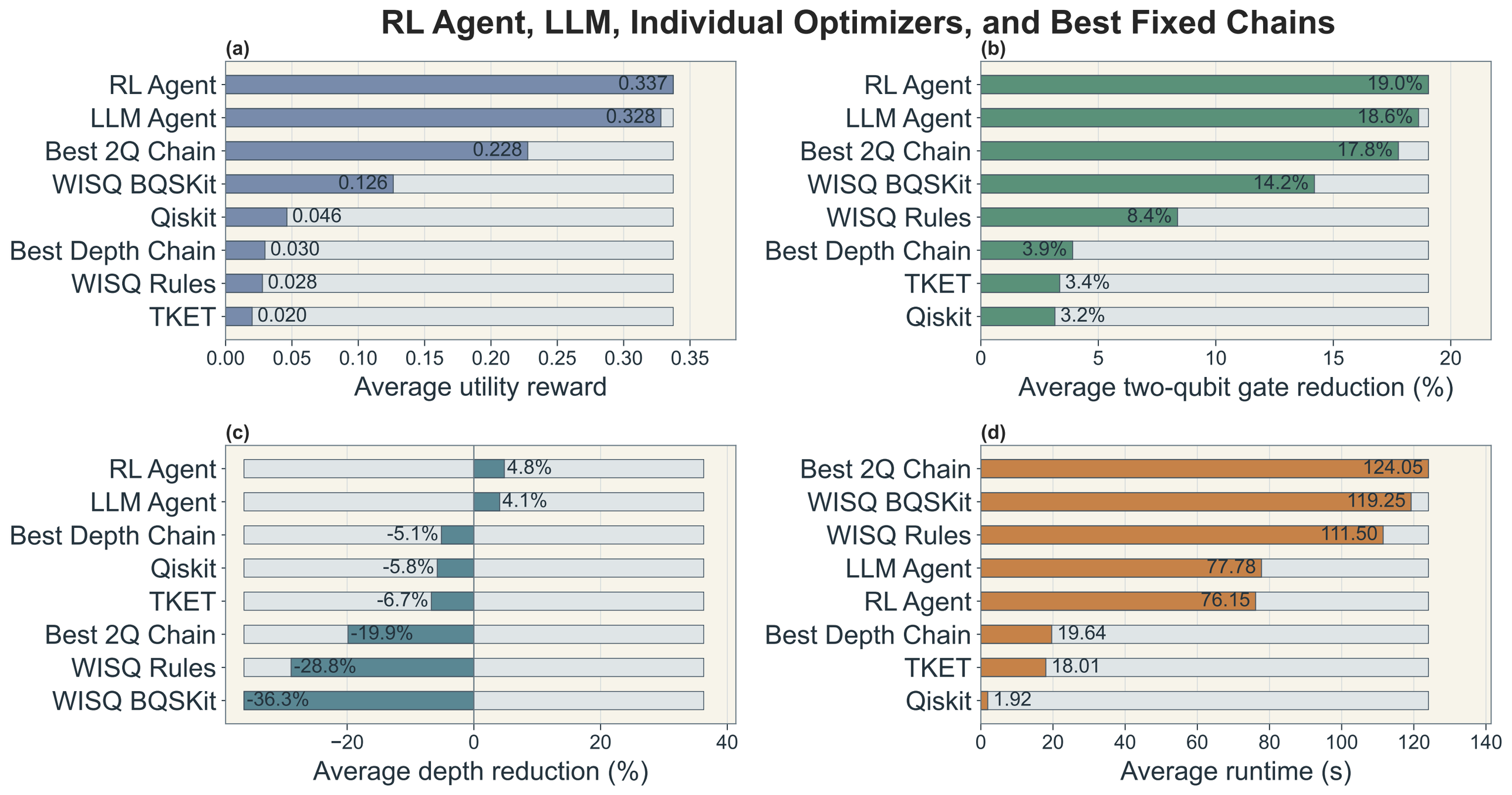
RL Agent, LLM, Individual Optimizers, and Best Fixed Chains
(a)
RL Agent
LLM Agent
Best 2Q Chain
WISQ BQSKit
Qiskit
Best Depth Chain
WISQ Rules
TKET
0.337
0.328
0.228
0.126
0.046
0.030
0.028
0.020
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Average utility reward
(b)
RL Agent
LLM Agent
Best 2Q Chain
WISQ BQSKit
WISQ Rules
Best Depth Chain
TKET
Qiskit
19.0%
18.6%
17.8%
14.2%
8.4%
3.9%
3.4%
3.2%
0
5
10
15
20
Average two-qubit gate reduction (%)
(c)
RL Agent
LLM Agent
Best Depth Chain
Qiskit
TKET
Best 2Q Chain
WISQ Rules
WISQ BQSKit
4.8%
4.1%
-5.1%
-5.8%
-6.7%
-19.9%
-28.8%
-36.3%
−20
0
20
40
Average depth reduction (%)
(d)
Best 2Q Chain
WISQ BQSKit
WISQ Rules
LLM Agent
RL Agent
Best Depth Chain
TKET
Qiskit
124.05
119.25
111.50
77.78
76.15
19.64
18.01
1.92
0
20
40
60
80
100
120
140
Average runtime (s)